

The problem:

Single agents

get overwhelmed

Introduction

What is a multi-agent system?

A multi-agent system uses multiple specialized agents working together instead of one agent trying to do everything.

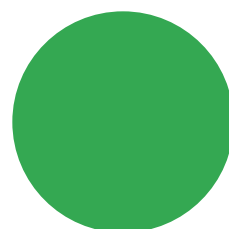
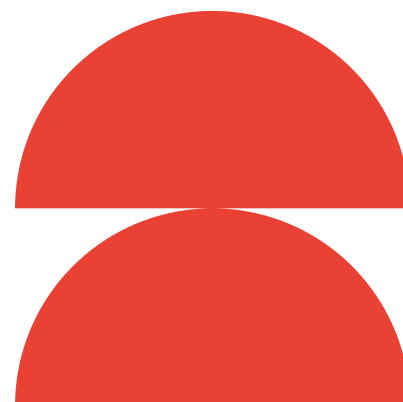
From ADK documentation

“Multi-agent systems in ADK allow you to build sophisticated applications where multiple specialized agents collaborate to solve complex problems. Rather than creating one monolithic agent, you can design a system where different agents handle different aspects of a task.”



Simple analogy:

Think of a customer service call center. Instead of one person handling billing, technical support, and account issues, you have specialists for each area. A coordinator routes calls to the right specialist.



The problem

Single agents get overwhelmed

Consider a customer service agent handling multiple areas:

```
Python
# One agent trying to do everything
customer_service = LlmAgent(
    model='gemini-2.5-flash',
    name='customer_service',
    instruction="""
    You handle:
    - Billing and payments
    - Technical troubleshooting
    - Account management
    - Product information
    """
)
```

When a single agent tries to handle too many responsibilities, several problems emerge. The agent's instruction becomes a lengthy list of different domains, making it difficult for the model to provide expert-level responses in any one area. This approach also creates maintenance challenges as the system grows.

What goes wrong:

- **Generic answers:** The agent knows a little about everything, but isn't expert in anything
- **Hard to maintain:** Adding new features means rewriting one giant instruction
- **Can't specialize:** No way to have different reasoning for different problem types